

AIM : Understand computer network fundamentals.

Basic Terms Used in Computer Networks

1. Network

A **network** is a collection of two or more computers or devices connected together so they can communicate and share data, files, printers, internet connections, and other resources.

Features

- Allows communication between devices.
- Enables file and resource sharing.
- Reduces cost by sharing hardware and software.
- Provides internet access to multiple devices.

Example

A college computer lab where all computers are connected to share files and use the same internet connection.

2. Node

A **node** is any device connected to a computer network that can send, receive, or forward data.

Types of Nodes

- Computer
- Laptop
- Smartphone
- Printer
- Router
- Switch
- Server

Example

If a network has 10 computers, 1 printer, and 1 router, then the network has **12 nodes**.

3. IP Address (Internet Protocol Address)

An **IP Address** is a unique numerical address assigned to every device connected to a network. It identifies the device and allows data to reach the correct destination.

Types of IP Address

A. IPv4

- Uses **32 bits**
- Written in four decimal numbers separated by dots.

Example:

192.168.1.10

B. IPv6

- Uses **128 bits**
- Created because IPv4 addresses are running out.

Example:

2001:0db8:85a3:0000:0000:8a2e:0370:7334

Advantages

- Identifies every device uniquely.
- Enables communication over the internet.

Example

Your laptop may have the IP address:

192.168.0.105

4. Router

A **router** is a networking device that connects different networks and forwards data packets between them. It chooses the best path for data to travel.

Functions

- Connects LAN to the Internet.
- Assigns IP addresses using DHCP.
- Directs traffic between networks.
- Provides Wi-Fi (wireless routers).

Advantages

- Connects multiple networks.
- Improves security using NAT and Firewall.
- Efficient routing of data.

Example

The Wi-Fi router installed at home connects all your devices to the internet.

5. Switch

A **switch** is a networking device used to connect multiple devices within the same Local Area Network (LAN). It sends data only to the intended device using MAC addresses.

Functions

- Connects computers in a LAN.
- Reduces unnecessary network traffic.
- Improves communication speed.

Advantages

- Faster than a hub.
- Reduces collisions.
- Secure communication.

Example

A school computer lab uses a switch to connect 30 computers.

6. Protocols

A **protocol** is a set of rules that defines how data is transmitted and received over a network.

Without protocols, computers cannot understand each other.

Common Network Protocols

Protocol Full Form		Purpose
HTTP	HyperText Transfer Protocol	Transfers web pages
HTTPS	HyperText Transfer Protocol Secure	Secure web browsing
FTP	File Transfer Protocol	File transfer
SMTP	Simple Mail Transfer Protocol	Sending emails

Protocol Full Form		Purpose
POP3	Post Office Protocol Version 3	Receiving emails
IMAP	Internet Message Access Protocol	Access emails from server
TCP	Transmission Control Protocol	Reliable data transmission
UDP	User Datagram Protocol	Fast data transmission
IP	Internet Protocol	Device addressing
DNS	Domain Name System	Converts website names into IP addresses

Example

When you open www.google.com, the DNS protocol converts the website name into an IP address so your browser can connect to Google's server.

7. Local Area Network (LAN)

A **Local Area Network (LAN)** is a network that connects devices within a small geographical area such as a home, school, office, or laboratory.

Characteristics

- Covers a small area.
- High speed.
- Low cost.
- Usually privately owned.

Example

A college computer lab with 50 connected computers.

8. Wide Area Network (WAN)

A **Wide Area Network (WAN)** connects multiple LANs over large geographical distances using public or private communication links.

Characteristics

- Covers cities, countries, or continents.
- Uses leased lines, satellites, or fiber optics.
- Slower than LAN.

- More expensive.

Example

A bank connecting all its branches across India.

9. Internet

The **Internet** is the world's largest network that connects millions of computers and devices across the globe.

It is often called the **Network of Networks** because it connects many smaller networks together.

Services Provided

- Web browsing
- Email
- Video conferencing
- Online gaming
- Cloud computing
- Social media
- Online banking

Advantages

- Instant communication.
- Access to unlimited information.
- Online education.
- E-commerce.
- Entertainment.

Example

Using Google, YouTube, WhatsApp, or Gmail requires the Internet.

10. Bandwidth

Bandwidth is the maximum amount of data that can be transmitted over a network in one second.

It determines how much data can travel through a network connection.

Units

- bps (bits per second)
- Kbps (Kilobits per second)
- Mbps (Megabits per second)
- Gbps (Gigabits per second)

Example

- 100 Mbps Internet connection
- 1 Gbps Fiber connection

Higher Bandwidth

- Faster downloads
- Better video streaming
- Smooth online gaming
- Supports more users simultaneously

Lower Bandwidth

- Slow downloads
- Video buffering
- Poor video calls
- Network congestion

Different Types of Computer Networks

Computer networks are classified based on the geographical area they cover and their purpose.

1. Local Area Network (LAN)

Definition

A **Local Area Network (LAN)** is a network that connects computers and other devices within a small geographical area such as a home, school, office, or laboratory.

Purpose

- Share files and folders.
- Share printers and scanners.
- Share a single Internet connection.

- Enable communication between nearby devices.
- Centralize data storage.

When is LAN Used?

- Schools and colleges
- Offices
- Computer laboratories
- Hospitals
- Homes

Characteristics

- Covers a small area (up to a few kilometers)
- High speed (100 Mbps to 10 Gbps)
- Low installation cost
- Easy to maintain

Advantages

- High data transfer speed
- Easy resource sharing
- Low maintenance cost
- Secure because it is privately managed

Disadvantages

- Limited geographical coverage
- Requires network administration
- Failure of the central device may affect the network

Example

A college computer lab where 40 computers are connected through a switch.

2. Wide Area Network (WAN)

Definition

A **Wide Area Network (WAN)** connects multiple LANs over large geographical areas such as cities, states, countries, or continents.

Purpose

- Connect distant offices or branches

- Enable global communication
- Share centralized databases
- Provide Internet connectivity across locations

When is WAN Used?

- Banks
- Government organizations
- Multinational companies
- Universities with multiple campuses

Characteristics

- Covers large distances
- Uses leased lines, satellites, fiber optics
- Slower than LAN
- Higher installation cost

Advantages

- Connects users worldwide
- Centralized management
- Supports remote work

Disadvantages

- Expensive
- Complex setup
- Higher latency

Example

A bank connecting all its branches across India.

3. Metropolitan Area Network (MAN)

Definition

A **Metropolitan Area Network (MAN)** connects multiple LANs within a city or metropolitan area.

Purpose

- Connect buildings in the same city
- Share resources among nearby offices

When is MAN Used?

- Universities
- City government offices
- Cable TV providers
- Internet Service Providers (ISPs)

Characteristics

- Covers 5–50 km
- Faster than WAN
- Larger than LAN

Advantages

- High-speed communication
- Connects multiple organizations
- Cost-effective for city-wide networks

Disadvantages

- More expensive than LAN
- Requires skilled management

Example

A university connecting different campuses within one city.

4. Personal Area Network (PAN)

Definition

A **Personal Area Network (PAN)** is a network connecting personal devices within a short range (typically up to 10 meters).

Purpose

- Connect personal electronic devices
- Transfer files
- Synchronize data

When is PAN Used?

- Connecting mobile phones
- Wireless headphones
- Smartwatches

- Bluetooth speakers

Characteristics

- Very small coverage area
- Uses Bluetooth, USB, NFC, or Wi-Fi Direct
- Low power consumption

Advantages

- Easy to set up
- Low cost
- Portable

Disadvantages

- Limited range
- Lower speed than LAN
- Supports only a few devices

Example

A smartphone connected to wireless earbuds using Bluetooth.

5. Wireless Local Area Network (WLAN)

Definition

A **Wireless Local Area Network (WLAN)** is a LAN that connects devices wirelessly using Wi-Fi instead of cables.

Purpose

- Provide wireless Internet access
- Enable mobility
- Reduce cable usage

When is WLAN Used?

- Homes
- Schools
- Cafes
- Airports
- Offices

Characteristics

- Uses Wi-Fi
- Flexible and portable
- Easy installation

Advantages

- No cables required
- Easy to expand
- Supports mobile devices

Disadvantages

- Lower security if not configured properly
- Signal interference
- Speed depends on distance and obstacles

Example

A home Wi-Fi network connecting laptops and smartphones.

6. Campus Area Network (CAN)

Definition

A **Campus Area Network (CAN)** connects multiple LANs within a campus or a large organization.

Purpose

- Connect different buildings within the same campus
- Share servers and Internet resources

When is CAN Used?

- Universities
- Military bases
- Corporate campuses
- Research centers

Characteristics

- Covers several buildings
- High-speed fiber connections
- Privately managed

Advantages

- High performance
- Easy management
- Secure communication

Disadvantages

- High installation cost
- Requires technical expertise

Example

A university connecting its library, laboratories, and administration building.

7. Storage Area Network (SAN)

Definition

A **Storage Area Network (SAN)** is a specialized high-speed network that connects servers to centralized storage devices.

Purpose

- Store large amounts of data
- Provide fast data access
- Improve backup and recovery

When is SAN Used?

- Data centers
- Banks
- Hospitals
- Cloud service providers

Characteristics

- Very high speed
- Dedicated storage network
- Highly reliable

Advantages

- Fast data access
- Centralized storage
- Easy backup

Disadvantages

- Expensive
- Complex to maintain

Example

A hospital storing patient records on centralized storage servers.

8. Virtual Private Network (VPN)

Definition

A **Virtual Private Network (VPN)** creates a secure, encrypted connection over the Internet, allowing users to access a private network remotely.

Purpose

- Secure online communication
- Protect sensitive data
- Enable remote access
- Improve privacy

When is VPN Used?

- Remote employees
- Online banking
- Secure public Wi-Fi usage
- Accessing company networks

Characteristics

- Uses encryption
- Secure communication
- Operates over the Internet

Advantages

- Protects privacy
- Secure remote access
- Encrypts data

Disadvantages

- Can reduce Internet speed
- Some VPN services are costly

- Requires proper configuration

Example

An employee working from home securely accessing the company's internal network.

9. Cloud Network

Definition

A **Cloud Network** is a network infrastructure hosted and managed by cloud service providers, allowing users to access computing resources over the Internet.

Purpose

- Host applications
- Store data
- Run virtual machines
- Provide scalable networking

When is Cloud Network Used?

- Cloud computing
- Online storage
- Web hosting
- Software-as-a-Service (SaaS)

Characteristics

- Accessible from anywhere
- Highly scalable
- Managed by cloud providers

Advantages

- Scalable
- Cost-effective
- High availability
- Easy remote access

Disadvantages

- Depends on Internet connectivity
- Possible security concerns if misconfigured
- Ongoing subscription costs

Example

Using cloud storage services such as Google Drive or Microsoft OneDrive.

Comparison of Network Types

Network Type	Coverage Area	Speed	Example
PAN	Up to 10 meters	Medium	Bluetooth devices
LAN	Home, office, school	Very High	Computer lab
WLAN	Wireless LAN	High	Home Wi-Fi
CAN	Campus	Very High	University campus
MAN	City	High	City-wide university network
WAN	Country/World	Medium	Bank branches
VPN	Internet-based secure connection	Depends on Internet	Remote office access
SAN	Data center	Very High	Enterprise storage
Cloud Network	Global (Internet)	Depends on Internet	Cloud services

Types of Network Architecture

Network architecture defines how devices communicate and share resources.

There are two main types:

1. Client-Server Architecture
2. Peer-to-Peer (P2P) Architecture

1. Client-Server Architecture

Definition

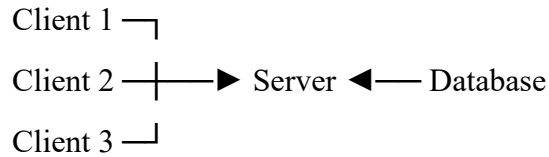
In a **Client-Server Architecture**, one or more powerful computers called **servers** provide services, data, or resources to multiple **client** computers.

Working

1. A client sends a request to the server.

2. The server processes the request.
3. The server sends the requested data or service back to the client.

Flow:



Example

- Web browsing (browser requests data from a web server)
- Gmail
- Banking systems
- College ERP systems

Advantages

- Centralized data management
- Better security
- Easy backup and maintenance
- Supports many users
- Easier software updates

Disadvantages

- Expensive to set up
- Server failure can affect all clients
- Requires skilled administrators
- Server can become a performance bottleneck if overloaded

2. Peer-to-Peer (P2P) Architecture

Definition

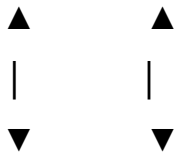
In a **Peer-to-Peer (P2P) Architecture**, every computer (peer) can act as both a client and a server. There is no central server.

Working

1. Each peer stores and shares its own resources.
2. A peer requests data directly from another peer.
3. Other peers can also request resources from it.

Flow:

Computer A $\longleftrightarrow$ Computer B



Computer C $\longleftrightarrow$ Computer D

Example

- File sharing between computers
- Small home networks
- BitTorrent file distribution
- Local multiplayer gaming

Advantages

- Low cost
- Easy to set up
- No dedicated server required
- No single point of failure

Disadvantages

- Lower security
- Difficult to manage as the network grows
- Data backup is decentralized
- Performance may decrease with many peers

Client-Server vs Peer-to-Peer Comparison

Feature	Client-Server	Peer-to-Peer
Central Server	Yes	No
Management	Centralized	Decentralized
Security	High	Lower
Cost	Higher	Lower
Scalability	High	Limited

Feature	Client-Server	Peer-to-Peer
Performance	Better for large networks	Suitable for small networks
Maintenance	Easier	More difficult
Example	Web servers, banking, email File sharing, small office networks	

Computer networking devices work at different layers of the **OSI Model** (Open Systems Interconnection Model). Each device performs specific functions based on the layer at which it operates.

Devices Used in Each OSI Layer

OSI Layer	Layer Name	Devices Used	Function
Layer 7	Application	Gateway, Proxy Server	Provides services to applications like web browsing, email, FTP
Layer 6	Presentation	Gateway, SSL/TLS Devices	Encrypts, decrypts, compresses, and formats data
Layer 5	Session	Gateway	Establishes, manages, and terminates communication sessions
Layer 4	Transport	Firewall, Load Balancer	Provides reliable data delivery and controls traffic
Layer 3	Network	Router, Layer 3 Switch	Uses IP addresses to route data between different networks
Layer 2	Data Link	Switch, Bridge, Network Interface Card (NIC), Wireless Access Point	Uses MAC addresses to forward data within the same network
Layer 1	Physical	Hub, Repeater, Modem (Physical signaling), Cables, Connectors	Transmits raw bits over physical media

Layer 1 – Physical Layer

Devices

- Hub
- Repeater
- Network Cable
- Fiber Optic Cable

- Connectors (RJ-45)
- Modem (physical signal conversion)

Function

- Transfers raw binary bits (0s and 1s)
- Defines electrical and mechanical standards
- Sends signals through cables

Example

A hub simply repeats incoming signals to every connected device.

Layer 2 – Data Link Layer

Devices

- Switch
- Bridge
- Network Interface Card (NIC)
- Wireless Access Point (WAP)

Function

- Uses **MAC Address**
- Detects errors
- Controls access to the network
- Delivers frames inside the same LAN

Example

A switch sends data only to the computer whose MAC address matches the destination.

Layer 3 – Network Layer

Devices

- Router
- Layer 3 Switch

Function

- Uses **IP Address**
- Finds the best path

- Routes packets between different networks

Example

A home router sends your request from your local network to the Internet.

Layer 4 – Transport Layer

Devices

- Firewall
- Load Balancer

Function

- Uses **TCP** and **UDP**
- Ensures reliable data delivery
- Performs flow control and error recovery
- Balances traffic between servers

Example

A firewall can allow or block traffic based on TCP or UDP port numbers.

Layer 5 – Session Layer

Devices

- Gateway

Function

- Creates, manages, and ends communication sessions
- Maintains synchronization between devices

Example

A video call session is established and maintained until you disconnect.

Layer 6 – Presentation Layer

Devices

- Gateway
- SSL/TLS Encryption Devices

Function

- Data encryption and decryption
- Compression
- Data format conversion

Example

HTTPS encrypts your banking information before it is transmitted.

Layer 7 – Application Layer**Devices**

- Gateway
- Proxy Server

Function

- Provides services directly to user applications
- Supports protocols such as HTTP, FTP, SMTP, and DNS

Example

When you open a website in a browser, communication begins at the Application Layer.

Common Networking Devices and Their OSI Layers

Device	OSI Layer	Address Used
Hub	Layer 1 (Physical)	No Address
Repeater	Layer 1 (Physical)	No Address
Cable	Layer 1 (Physical)	No Address
Bridge	Layer 2 (Data Link)	MAC Address
Switch	Layer 2 (Data Link)	MAC Address
Wireless Access Point	Layer 2 (Data Link)	MAC Address
NIC	Layer 2 (Data Link)	MAC Address
Router	Layer 3 (Network)	IP Address
Layer 3 Switch	Layer 3 (Network)	IP Address

Device	OSI Layer	Address Used
Firewall	Layer 3 / Layer 4 (sometimes Layer 7 for advanced firewalls)	IP Address, Port Number
Load Balancer	Layer 4 or Layer 7	Port Number / Application Data
Gateway	Layers 4–7 (can span multiple layers)	Depends on protocol
Proxy Server	Layer 7 (Application)	Application Data
Modem	Layer 1 (primarily)	No Network Address

Easy Trick to Remember

Layer	Device	Memory Trick
Layer 1	Hub, Repeater, Cable	Bits travel
Layer 2	Switch, Bridge, NIC, WAP	MAC Address
Layer 3	Router	IP Address
Layer 4	Firewall, Load Balancer	Port Numbers (TCP/UDP)
Layers 5–7	Gateway, Proxy	Applications and Services

One-Line Memory Formula

Hub → Switch → Router → Firewall → Gateway → Proxy

Types of Network Topologies

A **network topology** is the physical or logical arrangement of computers, cables, and networking devices in a network. It determines how devices are connected and how data flows between them.

1. Point-to-Point Topology

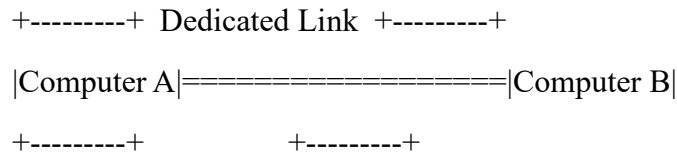
Definition

A **Point-to-Point Topology** is the simplest network topology in which **two devices are directly connected** by a communication link.

Working

- A dedicated communication link is established between two devices.
- Data travels directly from one device to the other.
- No intermediate device is required.

Diagram



Advantages

- Simple and easy to install.
- High data transfer speed.
- Secure because only two devices communicate.
- Easy to troubleshoot.

Disadvantages

- Connects only two devices.
- Not suitable for large networks.
- Requires additional links for more devices.

Example

- Two computers connected using an Ethernet cable.
- Communication between two routers using a leased line.

2. Mesh Topology

Definition

A **Mesh Topology** is a network where every device is connected to every other device through dedicated communication links.

Working

- Each node has a direct connection to every other node.
- If one connection fails, data can travel through another path.
- Can be **Full Mesh** (every device connected to every other device) or **Partial Mesh** (only selected devices have multiple connections).

Diagram

A

/\

/|\

B---+---C

\|/

\|/

D

Advantages

- Very reliable.
- High fault tolerance.
- No single point of failure.
- High security due to dedicated links.

Disadvantages

- Very expensive.
- Requires many cables.
- Difficult to install and maintain.
- Complex for large networks.

Example

- Military communication systems.
- Internet backbone.
- Banking networks.

3. Star Topology

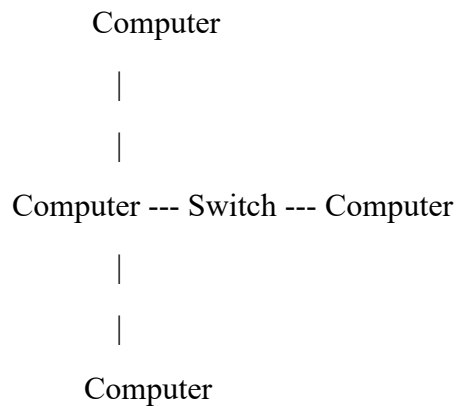
Definition

A **Star Topology** is a network where all devices are connected to a **central device**, usually a switch or hub.

Working

- Every computer has a separate cable connected to the central switch.
- Devices communicate through the central device.
- The switch forwards data to the intended destination.

Diagram



Advantages

- Easy to install.
- Easy to troubleshoot.
- Failure of one cable does not affect others.
- Easy to add new devices.
- High performance.

Disadvantages

- Central switch failure causes the entire network to stop.
- Requires more cable than a bus topology.
- Installation cost is higher.

Example

- Office networks.
- School computer labs.
- Home LANs.

4. Bus Topology

Definition

A **Bus Topology** uses a **single backbone cable** to which all devices are connected.

Working

- All devices share the same communication cable.
- Data is broadcast to all devices.
- Only the destination device accepts the data.

- Terminators are placed at both ends of the backbone cable to prevent signal reflection.

Diagram

Terminator

|

Computer -- Computer -- Computer -- Computer

|

Terminator

Advantages

- Low installation cost.
- Uses less cable.
- Easy to set up for small networks.

Disadvantages

- Backbone cable failure stops the entire network.
- Difficult to troubleshoot.
- Performance decreases as more devices are added.
- Data collisions may occur.

Example

- Small office networks (older Ethernet networks).

5. Ring Topology

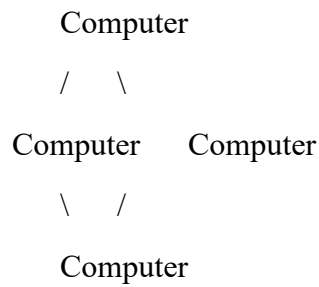
Definition

A **Ring Topology** connects each device to exactly **two neighboring devices**, forming a closed circular path.

Working

- Data travels around the ring.
- Each device receives and forwards data to the next device.
- Some ring networks use a **token** to control communication and avoid collisions.

Diagram



Advantages

- No data collisions in token-based systems.
- Equal access for all devices.
- Better performance than bus topology under heavy traffic.

Disadvantages

- Failure of one device or cable may stop the entire network (unless a dual ring is used).
- Difficult to expand.
- Troubleshooting can be challenging.

Example

- Fiber Distributed Data Interface (FDDI).
- Token Ring networks.

6. Tree Topology

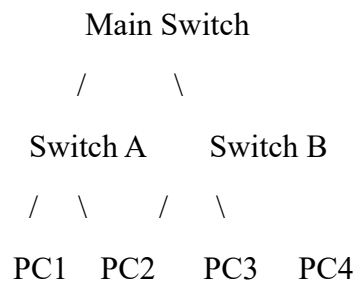
Definition

A **Tree Topology** combines **Star** and **Bus** topologies in a hierarchical structure.

Working

- Multiple star networks are connected through a central backbone.
- The root node connects to intermediate switches, which connect to end devices.
- Data travels from the root through branches to the destination.

Diagram



Advantages

- Easy to expand.
- Suitable for large organizations.
- Faults are easier to isolate.
- Supports hierarchical management.

Disadvantages

- Backbone failure affects many devices.
- More expensive than star topology.
- Installation is complex.

Example

- University campuses.
- Corporate offices.
- Large enterprise networks.

7. Hybrid Topology

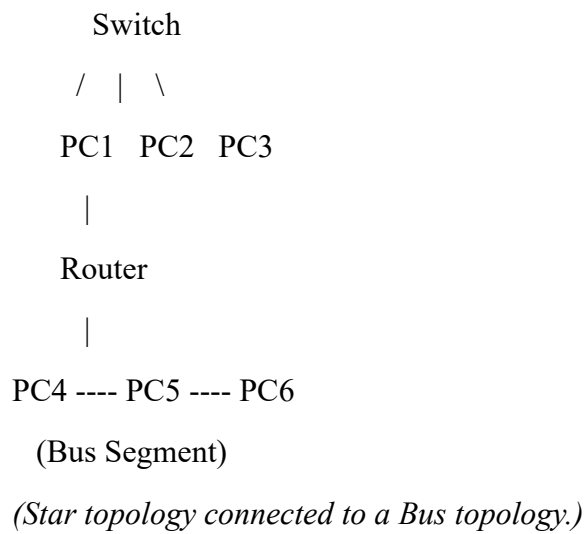
Definition

A **Hybrid Topology** combines two or more different network topologies to create a flexible and efficient network.

Working

- Different sections of the network use different topologies.
- These sections are connected through routers or switches.
- Data follows the rules of the respective topology within each section.

Diagram



Advantages

- Highly flexible.
- Scalable.
- Reliable.
- Fault isolation is easier.
- Can be customized according to organizational needs.

Disadvantages

- Expensive.
- Complex design.
- Requires skilled administrators.
- Maintenance is difficult.

Example

- Large companies.
- Hospitals.
- Universities.
- Data centers.

Comparison of Network Topologies

Topology	Structure	Cost	Reliability	Expansion	Example
Point-to-Point	Two devices directly connected	Low	High	Very Low	Router-to-router link
Mesh	Every device connected to every other device	Very High	Very High	Difficult	Military networks
Star	All devices connected to a central switch	Medium	High	Easy	Office LAN
Bus	Single backbone cable	Low	Low	Difficult	Small LAN
Ring	Circular connection	Medium	Medium	Difficult	Token Ring
Tree	Hierarchical (Star + Bus)	High	High	Easy	University campus
Hybrid	Combination of multiple topologies	Very High	Very High	Very Easy	Enterprise networks

Network Protocols

A **network protocol** is a set of rules and standards that allows devices to communicate with each other over a network. Protocols define how data is transmitted, received, and interpreted.

1. HTTP (HyperText Transfer Protocol)

Definition

HTTP is an application-layer protocol used to transfer web pages and other resources between a web browser and a web server.

How It Works

1. User enters a website URL in the browser.
2. The browser sends an **HTTP Request** to the web server.
3. The server processes the request.
4. The server sends an **HTTP Response** containing the requested webpage.
5. The browser displays the webpage.

Working Diagram

User → Browser → HTTP Request → Web Server

User ← Browser ← HTTP Response ← Web Server

Example

Opening <http://example.com> in a web browser.

Advantages

- Simple and fast.
- Supported by all web browsers.
- Easy to implement.

Disadvantages

- Data is sent in plain text.
- Not secure for sensitive information.

2. TCP (Transmission Control Protocol)

Definition

TCP is a transport-layer protocol that provides **reliable, connection-oriented** communication.

How It Works

1. TCP establishes a connection (Three-Way Handshake).
2. Data is divided into packets.
3. Packets are delivered in order.
4. Lost packets are retransmitted.
5. Connection is terminated after communication.

Working Diagram

Client ----- SYN -----> Server

Client <--- SYN-ACK ----- Server

Client ----- ACK -----> Server

Connection Established

Example

- Web browsing (HTTPS)
- Email

- Online banking

Advantages

- Reliable.
- Error checking.
- Ordered delivery.
- Data retransmission.

Disadvantages

- Slower than UDP.
- More overhead.

3. UDP (User Datagram Protocol)

Definition

UDP is a transport-layer protocol that provides **fast, connectionless** communication.

How It Works

1. Sender sends data packets.
2. No connection is established.
3. No acknowledgment is received.
4. Lost packets are not retransmitted.

Working Diagram

Sender -----> Receiver

(No Handshake)

Example

- Video streaming
- Online gaming
- Voice calls (VoIP)
- DNS queries

Advantages

- Very fast.
- Low overhead.
- Suitable for real-time applications.

Disadvantages

- Unreliable.
- No error recovery.
- Packets may arrive out of order.

4. BGP (Border Gateway Protocol)

Definition

BGP is the routing protocol used to exchange routing information between different autonomous systems (AS) on the Internet.

How It Works

1. Routers exchange routing tables.
2. BGP selects the best available route.
3. Data is forwarded using the selected path.

Working Diagram

ISP A ---- BGP ---- ISP B ---- BGP ---- ISP C

Example

Google's network communicating with an Internet Service Provider (ISP).

Advantages

- Scalable.
- Supports large networks.
- Efficient route selection.

Disadvantages

- Complex configuration.
- Slow convergence after network changes.

5. ARP (Address Resolution Protocol)

Definition

ARP maps an **IP address** to a **MAC address** within a local network.

How It Works

1. Device knows the destination IP address.
2. It broadcasts an ARP Request.

3. The destination device replies with its MAC address.
4. Communication begins using the MAC address.

Working Diagram

PC A

"Who has 192.168.1.5?"



PC B

"My MAC is AA:BB:CC:DD:EE:FF"

Example

A computer finding the MAC address of a printer on the same LAN.

Advantages

- Automatic address resolution.
- Fast communication in LAN.

Disadvantages

- Vulnerable to ARP spoofing attacks.
- Works only within a local network.

6. Internet Protocol (IP)

Definition

IP is a network-layer protocol responsible for assigning addresses to devices and routing packets between networks.

How It Works

1. Source device creates an IP packet.
2. Router forwards the packet.
3. Packet reaches the destination IP address.

Working Diagram

Computer → Router → Internet → Router → Destination

Example

Sending data from your laptop to a website.

Advantages

- Unique addressing.
- Supports routing.
- Scalable.

Disadvantages

- Does not guarantee delivery.
- Requires TCP for reliable communication.

7. DHCP (Dynamic Host Configuration Protocol)

Definition

DHCP automatically assigns IP addresses and other network settings to devices.

How It Works

1. Device sends DHCP Discover.
2. DHCP Server sends DHCP Offer.
3. Device sends DHCP Request.
4. DHCP Server sends DHCP Acknowledgment (ACK).

Working Diagram

Computer → Discover

Server → Offer

Computer → Request

Server → ACK

Example

Your phone automatically gets an IP address when connected to Wi-Fi.

Advantages

- Automatic configuration.
- Reduces manual work.
- Prevents IP conflicts.

Disadvantages

- DHCP server failure affects new devices.
- Depends on server availability.

8. ICMP (Internet Control Message Protocol)

Definition

ICMP is used for error reporting and network diagnostics.

How It Works

1. Device sends an ICMP Echo Request.
2. Destination replies with an Echo Reply.
3. Network administrators use the result to test connectivity.

Working Diagram

Computer ---- Ping ----> Server

Computer <--- Reply --- Server

Example

Using the ping command to check whether a server is reachable.

Advantages

- Network troubleshooting.
- Error reporting.
- Simple diagnostics.

Disadvantages

- Can be blocked by firewalls.
- Can be misused in some network attacks.

9. SNMP (Simple Network Management Protocol)

Definition

SNMP is used to monitor and manage network devices such as routers, switches, and printers.

How It Works

1. SNMP Manager requests information.
2. Network device (Agent) responds with status.
3. Administrator monitors device performance.

Working Diagram

SNMP Manager <-----> Router/Switch/Printer

Example

Monitoring CPU usage of a router.

Advantages

- Centralized monitoring.
- Easy device management.
- Fault detection.

Disadvantages

- Older versions have weaker security.
- Requires proper configuration.

10. FTP (File Transfer Protocol)

Definition

FTP is an application-layer protocol used to transfer files between computers over a network.

How It Works

1. Client connects to FTP Server.
2. User logs in.
3. Files are uploaded or downloaded.
4. Connection is closed.

Working Diagram

FTP Client <-----> FTP Server

Example

Uploading website files to a web hosting server.

Advantages

- Fast file transfer.
- Supports large files.
- Easy file management.

Disadvantages

- Data is not encrypted by default.
- Less secure than SFTP or FTPS.

11. POP3 (Post Office Protocol Version 3)

Definition

POP3 is used to **receive emails** from an email server to a local device.

How It Works

1. Email client connects to the mail server.
2. Downloads emails.
3. Emails are usually removed from the server after download.
4. User reads emails offline.

Working Diagram

Mail Server ---- POP3 ----> User's Computer

Example

Microsoft Outlook downloading emails from a mail server.

Advantages

- Offline email access.
- Saves server storage.
- Simple to use.

Disadvantages

- Emails are often removed from the server.
- Not suitable for accessing the same mailbox from multiple devices.

12. TELNET

Definition

TELNET is an application-layer protocol used for remote login and command-line access to another computer over a network.

How It Works

1. User starts a Telnet client.
2. Connects to the remote server.
3. User enters commands.
4. Server executes the commands and returns the output.

Working Diagram

User Computer ---- TELNET ----> Remote Server

Example

A network administrator remotely managing a router or Linux server.

Advantages

- Simple remote administration.
- Easy to configure.
- Lightweight.

Disadvantages

- No encryption.
- Usernames and passwords are sent in plain text.
- Largely replaced by SSH for secure remote access.

Comparison Table

Protocol	Full Form	Layer	Main Purpose	Example
HTTP	HyperText Transfer Protocol	Application	Load web pages	Browsing websites
TCP	Transmission Control Protocol	Transport	Reliable communication	Online banking
UDP	User Datagram Protocol	Transport	Fast communication	Online gaming
BGP	Border Gateway Protocol	Network	Internet routing	ISP routing
ARP	Address Resolution Protocol	Data Link	IP to MAC mapping	Finding printer MAC address
IP	Internet Protocol	Network	Addressing and routing	Sending data over the Internet
DHCP	Dynamic Host Configuration Protocol	Application	Automatic IP assignment	Connecting to Wi-Fi
ICMP	Internet Control Message Protocol	Network	Diagnostics and error reporting	ping command
SNMP	Simple Network Management Protocol	Application	Device monitoring	Router monitoring
FTP	File Transfer Protocol	Application	File transfer	Uploading website files

Protocol	Full Form	Layer	Main Purpose	Example
POP3	Post Office Protocol Version 3	Application	Receiving emails	Downloading emails
TELNET	Telecommunication Network	Application	Remote login	Remote server administration

Easy Memory Trick

- **HTTP** → Opens websites.
- **TCP** → Reliable delivery.
- **UDP** → Fast delivery.
- **BGP** → Routes traffic between ISPs.
- **ARP** → Finds MAC address from an IP address.
- **IP** → Gives every device an address and routes packets.
- **DHCP** → Automatically assigns IP addresses.
- **ICMP** → Checks connectivity (ping).
- **SNMP** → Monitors network devices.
- **FTP** → Transfers files.
- **POP3** → Downloads emails.
- **TELNET** → Remotely controls another computer (without encryption).

IP Address

Definition

An **IP (Internet Protocol) Address** is a **unique numerical address** assigned to every device connected to a network. It identifies the device and allows data to be sent to the correct destination.

Just as every house has a unique postal address, every device on a network has a unique IP address.

Purpose of an IP Address

- Identifies devices on a network.
- Enables communication between devices.
- Helps routers deliver data to the correct destination.
- Supports Internet and local network connectivity.

Example

- Laptop: **192.168.1.10**
- Smartphone: **192.168.1.15**
- Printer: **192.168.1.20**

IPv4 (Internet Protocol Version 4)

Definition

IPv4 is the fourth version of the Internet Protocol and is the most widely used IP addressing system. It uses **32-bit addresses**, allowing approximately **4.3 billion unique IP addresses**.

Format

IPv4 consists of **four decimal numbers (octets)** separated by dots.

Example

192.168.10.25

Each number ranges from **0 to 255**.

Example:

172.16.0.1

10.0.0.15

8.8.8.8

Structure

192 . 168 . 1 . 25

| | | |

Octet Octet Octet Octet

Each octet contains **8 bits**.

$8 + 8 + 8 + 8 = 32$ bits

Features

- 32-bit address.
- Approximately **4.3 billion addresses**.
- Easy to read.
- Widely supported.
- Uses decimal notation.

Advantages

- Simple and widely used.

- Easy to configure.
- Compatible with almost all devices.

Disadvantages

- Limited number of addresses.
- Address exhaustion due to the growth of Internet users.
- Requires techniques like NAT (Network Address Translation) to conserve addresses.

IPv6 (Internet Protocol Version 6)

Definition

IPv6 is the latest version of the Internet Protocol. It uses **128-bit addresses**, providing an extremely large address space to support the growing number of Internet-connected devices.

Format

IPv6 consists of **eight groups** of four hexadecimal digits separated by colons.

Example

2001:0db8:85a3:0000:0000:8a2e:0370:7334

It can also be written in a shortened form:

2001:db8:85a3::8a2e:370:7334

Structure

2001:0db8:85a3:0000:0000:8a2e:0370:7334

Each block contains **16 bits**.

$$16 \times 8 = 128 \text{ bits}$$

Features

- 128-bit address.
- Uses hexadecimal numbers (0–9 and A–F).
- Nearly unlimited number of IP addresses.
- Built-in security support through IPsec.
- Better routing efficiency.

Advantages

- Huge address space.
- Better security.
- No need for NAT in most deployments.

- Improved performance for modern networks.
- Supports automatic address configuration.

Disadvantages

- More difficult to remember due to its length.
- Migration from IPv4 requires planning.
- Some older devices may not support IPv6.

Difference Between IPv4 and IPv6

Feature	IPv4	IPv6
Full Form	Internet Protocol Version 4	Internet Protocol Version 6
Address Length	32 bits	128 bits
Address Format	Decimal numbers	Hexadecimal numbers
Example	192.168.1.10	2001:db8::1
Number of Addresses	About 4.3 billion	About 340 undecillion (3.4×10^{38})
Notation	Dot (.)	Colon (:))
Address Blocks	4 octets	8 groups
NAT Required	Yes, often used	Usually not required
Security	Optional	Built-in IPsec support
Auto Configuration	Limited	Supported (Stateless Address Autoconfiguration)
Speed	Good	Better routing efficiency
Current Usage	Still widely used	Increasing adoption worldwide

IPv4 vs IPv6 Examples

IPv4	IPv6
192.168.1.1	2001:db8::1
10.0.0.5	fe80::1

IPv4

IPv6

172.16.1.100 2404:6800:4003:c00::64

Working Example

Suppose you want to open a website.

Using IPv4

Laptop (192.168.1.10)

|



Router

|



Internet

|



Website Server (142.250.183.78)

Using IPv6

Laptop (2001:db8::10)

|



Router

|



Internet

|



Website Server (2607:f8b0:4005:805::200e)

In both cases, routers use the IP address to forward data to the correct destination.

Easy Memory Trick

- **IPv4**
 - **4 parts**
 - **32 bits**
 - **Dots (.)**
 - Example: 192.168.1.10
- **IPv6**
 - **8 groups**
 - **128 bits**
 - **Colons (:)**
 - Example: 2001:db8::1

Quick Revision Table

Topic	IPv4	IPv6
Bits	32	128
Groups	4	8
Separator	Dot (.)	Colon (:)
Format	Decimal	Hexadecimal
Address Capacity ~4.3 Billion ~340 Undecillion		
Security	Basic	Better (IPsec support)

Network Security

Definition

Network Security is the practice of protecting a computer network, devices, and data from unauthorized access, cyberattacks, misuse, theft, and damage.

The main goal of network security is to ensure:

- **Confidentiality** – Only authorized users can access data.
- **Integrity** – Data is not altered or modified without permission.
- **Availability** – Network services remain available to authorized users.

1. Firewalls

Definition

A **Firewall** is a network security device or software that monitors and controls incoming and outgoing network traffic based on predefined security rules.

It acts as a **barrier** between a trusted internal network and an untrusted external network (such as the Internet).

Purpose

- Prevent unauthorized access.
- Block malicious traffic.
- Protect computers and servers.
- Monitor network communication.
- Improve network security.

Functionality

1. Data packets enter the firewall.
2. The firewall checks each packet against security rules.
3. Safe traffic is allowed.
4. Suspicious or unauthorized traffic is blocked.

Working Diagram

Internet

|



+-----+

| Firewall |

+-----+

| |

Allowed Blocked

|

Internal Network

Types of Firewalls

A. Packet Filtering Firewall

- Examines each packet individually.
- Uses IP addresses, port numbers, and protocols.
- Fast and simple.

Example: Home router firewall.

B. Stateful Inspection Firewall

- Tracks active connections.
- Allows only packets belonging to valid sessions.
- More secure than packet filtering.

Example: Enterprise firewalls.

C. Proxy Firewall (Application-Level Firewall)

- Acts as an intermediary between users and servers.
- Inspects application-layer traffic.
- Hides internal IP addresses.

Example: Web proxy server.

D. Next-Generation Firewall (NGFW)

- Combines traditional firewall features with advanced security.
- Detects malware and application-level attacks.
- Supports Intrusion Prevention Systems (IPS).

Example: Enterprise security gateways.

Advantages

- Blocks unauthorized access.
- Protects against cyberattacks.
- Monitors network traffic.
- Improves overall security.

Disadvantages

- Incorrect configuration may block valid traffic.
- Advanced firewalls are expensive.
- Cannot stop attacks originating inside the network.

2. Encryption

Definition

Encryption is the process of converting readable data (**plaintext**) into unreadable data (**ciphertext**) using an encryption algorithm and a key.

Only users with the correct decryption key can read the original data.

Purpose

- Protect confidential information.
- Prevent unauthorized access.
- Secure online communication.
- Protect stored data.

Functionality

1. Plaintext is entered.
2. An encryption algorithm converts it into ciphertext.
3. Ciphertext is transmitted or stored.
4. The receiver decrypts it using the correct key.

Working Diagram

Plain Text

|

Encryption

|

Cipher Text

|

Internet

|
Decryption
|
Original Message

Types of Encryption

A. Symmetric Encryption

- Same key is used for encryption and decryption.
- Faster.
- Suitable for encrypting large amounts of data.

Examples

- AES
- DES

B. Asymmetric Encryption

- Uses two keys:
 - Public Key
 - Private Key
- More secure than symmetric encryption.

Examples

- RSA
- ECC

Advantages

- Protects sensitive data.
- Ensures privacy.
- Prevents data theft.
- Secures Internet communication.

Disadvantages

- Key management can be challenging.
- Asymmetric encryption is slower.
- Lost encryption keys may make data inaccessible.

3. Access Control

Definition

Access Control is a security mechanism that determines **who can access network resources and what actions they are allowed to perform.**

Purpose

- Prevent unauthorized access.
- Protect sensitive information.
- Control user permissions.
- Improve security.

Functionality

1. User attempts to log in.
2. System verifies identity (authentication).
3. Permissions are checked (authorization).
4. Access is granted or denied.

Working Diagram

User

|

Login

|

Authentication

|

Authorization

|

Access Granted / Access Denied

Types of Access Control

A. Discretionary Access Control (DAC)

- Resource owner decides who gets access.

Example: Sharing a file with selected users.

B. Mandatory Access Control (MAC)

- Access is controlled by the system using security classifications.
- Common in military and government organizations.

C. Role-Based Access Control (RBAC)

- Permissions are assigned based on user roles.

Example

- Student → View records
- Teacher → Edit marks
- Admin → Full access

D. Attribute-Based Access Control (ABAC)

- Access depends on attributes such as:
 - User role
 - Department
 - Location
 - Time of access

Advantages

- Prevents unauthorized access.
- Protects confidential information.
- Simplifies permission management.
- Improves accountability.

Disadvantages

- Requires proper configuration.
- Complex in large organizations.
- Misconfigured permissions can cause security issues.

4. Intrusion Detection System (IDS)

Note: The correct term is **Intrusion Detection System (IDS)**, not "Intruition Defection."

Definition

An **Intrusion Detection System (IDS)** monitors network traffic and system activities to detect suspicious behavior, attacks, or security policy violations.

Unlike a firewall, an IDS **detects and alerts** but usually **does not block** the attack.

Purpose

- Detect cyberattacks.
- Identify unauthorized access.
- Monitor network activity.
- Alert administrators about suspicious behavior.

Functionality

1. Network traffic is monitored.
2. IDS analyzes packets and events.
3. Suspicious activity is detected.
4. Alerts are generated for administrators.

Working Diagram

Internet

|

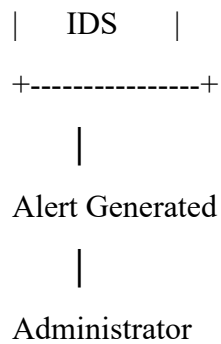


Network Traffic

|



+-----+



Types of IDS

A. Network-Based IDS (NIDS)

- Monitors the entire network.
- Detects attacks traveling across the network.

Example: Monitoring traffic entering an office network.

B. Host-Based IDS (HIDS)

- Installed on individual computers or servers.
- Monitors files, processes, and system logs.

Example: Detecting unauthorized changes to a server.

C. Signature-Based IDS

- Detects known attack patterns.
- Compares traffic against a database of attack signatures.

Advantage: Accurate for known threats.

Disadvantage: Cannot detect new or unknown attacks.

D. Anomaly-Based IDS

- Learns normal network behavior.
- Detects unusual or abnormal activities.

Advantage: Can detect unknown attacks.

Disadvantage: May generate false alarms.

Advantages

- Detects attacks early.
- Improves network monitoring.
- Generates security alerts.
- Helps investigate incidents.

Disadvantages

- Cannot automatically stop most attacks.
- May produce false positives.
- Requires regular updates and monitoring.

Comparison Table

Security Mechanism	Purpose	Functionality	Common Types
Firewall	Blocks unauthorized network access	Filters incoming and outgoing traffic	Packet Filtering, Stateful Inspection, Proxy, Next-Generation Firewall (NGFW)
Encryption	Protects data confidentiality	Converts plaintext into ciphertext	Symmetric, Asymmetric
Access Control	Restricts access to resources	Authenticates users and authorizes permissions	DAC, MAC, RBAC, ABAC
Intrusion Detection System (IDS)	Detects suspicious activities	Monitors traffic and generates alerts	NIDS, HIDS, Signature-Based, Anomaly-Based

Quick Revision

Topic	Purpose	Example
Firewall	Block unwanted network traffic	Office firewall
Encryption	Protect data by converting it into unreadable form	HTTPS, Online Banking
Access Control	Decide who can access resources	User login with permissions
IDS	Detect attacks and generate alerts	Monitoring network traffic

CONCLUSION : CASE STUDY UNDERSTOOD SUCCESSFULLY.